

Challenge (Habanero)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{3}}$.

- (A) $\frac{\sqrt{3}}{3}$
- (B) $\frac{1}{3}$
- (C) $\sqrt{3}$
- (D) $\frac{1}{\sqrt{3}}$
- (E) $\frac{3}{\sqrt{3}}$

Q2. In a gaming tournament, the average score is $\frac{1}{\sqrt{2}}$ points per player. Rationalize the denominator.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2}$
- (C) $\sqrt{2}$
- (D) $\frac{1}{2\sqrt{2}}$
- (E) $\frac{2}{\sqrt{2}}$

Q3. A data packet has a size of $\frac{1}{\sqrt{5}}$ megabytes. Rationalize the denominator.

- (A) $\frac{\sqrt{5}}{5}$
- (B) $\frac{1}{5}$
- (C) $\sqrt{5}$
- (D) $\frac{1}{5\sqrt{5}}$
- (E) $\frac{5}{\sqrt{5}}$

Q4. Simplify $\frac{\sqrt{6}}{\sqrt{2}}$ by rationalizing the denominator.

- (A) $\sqrt{3}$
- (B) $\frac{\sqrt{3}}{2}$
- (C) $\sqrt{2}$
- (D) $\frac{\sqrt{2}}{3}$
- (E) $\frac{\sqrt{6}}{2}$

Q5. A gamer's score is $\frac{1}{\sqrt{10}}$ points per minute. Rationalize the denominator.

- (A) $\frac{\sqrt{10}}{10}$
- (B) $\frac{1}{10}$
- (C) $\sqrt{10}$
- (D) $\frac{1}{10\sqrt{10}}$
- (E) $\frac{10}{\sqrt{10}}$

Q6. Rationalize the denominator of $\frac{\sqrt{3}}{\sqrt{6}}$.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2}$
- (C) $\frac{\sqrt{3}}{3}$
- (D) $\frac{\sqrt{6}}{6}$
- (E) $\frac{\sqrt{2}}{6}$

Q7. Simplify $\frac{1}{\sqrt{15}}$ by rationalizing the denominator.

- (A) $\frac{\sqrt{15}}{15}$
- (B) $\frac{1}{15}$
- (C) $\sqrt{15}$
- (D) $\frac{1}{15\sqrt{15}}$
- (E) $\frac{15}{\sqrt{15}}$

Q8. A coding logic problem has a complexity of $\frac{1}{\sqrt{8}}$ hours. Rationalize the denominator.

- (A) $\frac{\sqrt{2}}{4}$
- (B) $\frac{1}{4}$
- (C) $\sqrt{2}$
- (D) $\frac{1}{4\sqrt{2}}$
- (E) $\frac{4}{\sqrt{2}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{\sqrt{12}}{\sqrt{8}}$ and simplify. Show all steps.

Q10. A scoreboard displays the average score as $\frac{1}{\sqrt{20}}$ points per game. Rationalize the denominator and simplify. Explain the process.

Chef's Tips

- Tip 1: Review the concept of rationalizing denominators
- Tip 2: Emphasize the importance of simplifying radicals
- Tip 3: Provide examples of real-world applications of rationalizing denominators